Permutations and Combinations

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Permutations & Combinations Practice Exercises**

Find the number of ways to arrange the 7 letters in the word SUCCESS.

The word SUCCESS has 7 letters with 3 S's, 2 C's, 1 U, and 1 E. The number of distinct permutations is $\frac{7!}{3!\,2!} = \frac{5040}{6 \times 2} = \frac{5040}{12} = 420$.

2 **[Foundational]** • **Target: Permutations & Combinations Practice Exercises**

In how many ways can 5 people be seated in a row if two particular people (A and B) must not sit next to each other?

Total $= 5! = 120$. Cases where A & B are adjacent: treat AB as a block $\rightarrow 4! \times 2 = 48$. Answer: $120 - 48 = 72$.

3 **[Foundational]** • **Target: Permutations & Combinations Practice Exercises**

The circle C has equation $x^2 + y^2 - 6x + 4y - 12 = 0$. (a) Find the coordinates of the centre and the exact radius of C. (b) Find the equation of the tangent to C at the point $P(6, 2)$.

(a) Complete the squares for x and y: $(x^2 - 6x + 9) + (y^2 + 4y + 4) = 12 + 9 + 4 \Rightarrow (x - 3)^2 + (y + 2)^2 = 25$ The centre is $(3,\ -2)$ and the radius is $\sqrt{25} = 5$. (b) The gradient of the radius from centre $(3,\ -2)$ to $P(6, 2)$ is: $m_{\text{rad}} = \frac{2 - (-2)}{6 - 3} = \frac{4}{3}$ The tangent is perpendicular to the radius, so its gradient is $m_{\tan} = -\frac{3}{4}$. Using point-slope form with $P(6, 2)$.

4 **[Foundational]** • **Target: Permutations & Combinations Practice Exercises**

Given vectors $\mathbf{a} = 2\mathbf{i} - 5\mathbf{j}$ and $\mathbf{b} = -3\mathbf{i} + 4\mathbf{j}$, find the scalar projection of $\mathbf{a}$ onto $\mathbf{b}$ and the angle θ between them to the nearest tenth of a degree.

Calculate the dot product: $\mathbf{a} \cdot \mathbf{b} = (2)(-3) + (-5)(4) = -6 - 20 = -26.$ Calculate the magnitudes: $|\mathbf{a}| = \sqrt{2^2 + (-5)^2} = \sqrt{4 + 25} = \sqrt{29} \approx 5.385,\ |\mathbf{b}| = \sqrt{(-3)^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5.$ Scalar projection of $\mathbf{a}$ onto $\mathbf{b}$: $\text{comp}_{\mathbf{b}}\mathbf{a} = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{b}|} = \frac{-26}{5} = -5.2.$ Angle θ between vectors.

5 **[Foundational]** • **Target: Permutations & Combinations Practice Exercises**

Find the equation of the normal to the curve $y = \frac{4}{\sqrt{x}} + 2x$ at the point where $x = 4$.

Rewrite in index form: $y = 4x^{-1/2} + 2x$. At $x = 4$: $y = 4(4^{-1/2}) + 2(4) = 4\left(\frac{1}{2}\right) + 8 = 2 + 8 = 10$. Point of contact is $(4, 10)$. Differentiate: $\frac{dy}{dx} = 4\left(-\frac{1}{2}x^{-3/2}\right) + 2 = -2x^{-3/2} + 2 = -\frac{2}{x^{3/2}} + 2$ At $x = 4$: $x^{3/2} = (\sqrt{4})^3 = 8$. Thus: $m_{\tan} = -\frac{2}{8} + 2 = -\frac{1}{4} + 2 = \frac{7}{4}$ The normal has gradient $m_{\text{norm}} = -\frac{1}{7/4} = -\frac{4}{7}$. Using point $(4, 10)$.

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6 [Foundational] • Target: Permutations & Combinations Practice Exercises

Solve the equation $2\cos 2x = 1 + \sin x$ for $0 \le x \le 2\pi$ radians, giving answers to 3 significant figures.

Substitute the double angle identity $\cos 2x = 1 - 2\sin^2 x$: $2(1 - 2\sin^2 x) = 1 + \sin x$ $2 - 4\sin^2 x = 1 + \sin x \Rightarrow 4\sin^2 x + \sin x - 1 = 0$ Apply quadratic formula for $\sin x$: $\sin x = \frac{-1 \pm \sqrt{1 - 4(4)(-1)}}{8} = \frac{-1 \pm \sqrt{17}}{8}$ Case 1: $\sin x = \frac{-1 + \sqrt{17}}{8} \approx 0.3904$: $x = 0.401$ rad, $\pi - 0.401 = 2.74$ rad Case 2: $\sin x = \frac{-1 - \sqrt{17}}{8} \approx -0.6404$.

7 [Foundational] • Target: Permutations & Combinations Practice Exercises

A block of mass 8 kg rests in limiting equilibrium on a rough plane inclined at 30° to the horizontal. Taking $g = 9.8$ m/s^2: (a) calculate the normal reaction R, (b) find the coefficient of friction μ between the block and the plane.

(a) Resolve perpendicular to the inclined plane: $R = mg\cos 30^\circ = 8(9.8)\cos 30^\circ = 78.4 \times \frac{\sqrt{3}}{2} = 39.2\sqrt{3} \approx 67.9$ N (3 s.f.) (b) Resolve parallel to the plane. Since the block is in limiting equilibrium on the verge of sliding down, friction $F = \mu R$ acts up the plane: $F = mg\sin 30^\circ = 8(9.8)\sin 30^\circ = 78.4 \times 0.5 = 39.2$ N Setting $F = \mu R$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Permutations & Combinations Practice Exercises

Find the constant term (independent of x) in the binomial expansion of $\left(2x^2 - \frac{1}{x}\right)^9$.

Step 1: Write the general term T_{k+1}: $T_{k+1} = \binom{9}{k}(2x^2)^{9-k}\left(-\frac{1}{x}\right)^k = \binom{9}{k}2^{9-k}x^{2(9-k)}(-1)^k x^{-k} = \binom{9}{k}2^{9-k}(-1)^k x^{18-3k}$. Step 2: Find k for the term independent of x: Set exponent equal to 0: $18 - 3k = 0 \Rightarrow 3k = 18 \Rightarrow k = 6$. Step 3: Evaluate coefficient at $k = 6$: $\binom{9}{6} = \binom{9}{3} = \frac{9 \times 8 \times 7}{3 \times 2 \times 1} = 84$. .

9 [Intermediate] • Target: Permutations & Combinations Practice Exercises

The curve C has equation $x^3 - 2xy + y^2 = 4$. Find the coordinates of the two points on C where the tangent is parallel to the y-axis (vertical).

Differentiate implicitly with respect to x: $3x^2 - 2\left(y + x\frac{dy}{dx}\right) + 2y\frac{dy}{dx} = 0$ $3x^2 - 2y + (2y - 2x)\frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = \frac{2y - 3x^2}{2y - 2x}$ A vertical tangent occurs when the denominator is zero (and numerator non-zero): $2y - 2x = 0 \Rightarrow y = x$ Substitute $y = x$ into the original curve equation: $x^3 - 2x(x) + x^2 = 4 \Rightarrow x^3 - x^2 - 4 = 0$ Test integer factors: $x = 2$ gives $8 - 4 - 4 = 0$. Factoring gives $(x - 2)(x^2 + x + 2) = 0$.

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10 [Intermediate] • Target: Permutations & Combinations Practice Exercises

A bag contains 5 red and 4 green marbles. Three marbles are drawn at random without replacement. Find the probability that: (a) all three are red, (b) at least two are red.

Total marbles = 9. Number of ways to choose 3 marbles = $\binom{9}{3} = \frac{9\times8\times7}{3\times2\times1} = 84$. (a) All 3 are red: $\binom{5}{3} = 10$. Probability: $P(\text{all red}) = \frac{10}{84} = \frac{5}{42} \approx 0.119$ (b) 'At least two red' means exactly 2 red and 1 green, or exactly 3 red: Ways for 2 red, 1 green: $\binom{5}{2} \times \binom{4}{1} = 10 \times 4 = 40$. Ways for 3 red: $\binom{5}{3} = 10$. Total favorable outcomes: $40 + 10 = 50$. Probability.

11 [Intermediate] • Target: Permutations & Combinations Practice Exercises

A train accelerates uniformly from rest to $30\ \text{m/s}$ in $20\ \text{s}$, travels at constant speed $30\ \text{m/s}$ for $60\ \text{s}$, and then decelerates uniformly to rest in $15\ \text{s}$. Find: (a) the acceleration in the first phase, (b) the total distance travelled, (c) the average speed for the entire journey.

(a) $a_1 = \frac{v-u}{t} = \frac{30-0}{20} = 1.5\ \text{m/s}^2$. (b) Total distance is the area of the trapezoid on the v-t graph with parallel sides $60\ \text{s}$ and $20 + 60 + 15 = 95\ \text{s}$, and height $30\ \text{m/s}$: $s = \frac{60+95}{2} \times 30 = \frac{155}{2} \times 30 = 155 \times 15 = 2325\ \text{m}$ (c) Average speed.

12 [Intermediate] • Target: Permutations & Combinations Practice Exercises

Convert the Cartesian equation $(x^2 + y^2)^2 = 4(x^2 - y^2)$ into polar form and identify the name and key features of the resulting curve.

Substitute standard Cartesian-to-polar relations $x^2 + y^2 = r^2$, $x = r\cos\theta$, and $y = r\sin\theta$: $(r^2)^2 = 4(r^2\cos^2\theta - r^2\sin^2\theta) \Rightarrow r^4 = 4r^2(\cos^2\theta - \sin^2\theta)$. Apply the double-angle identity $\cos^2\theta - \sin^2\theta = \cos(2\theta)$: $r^4 = 4r^2\cos(2\theta)$. Divide through by r^2 (for $r \neq 0$): $r^2 = 4\cos(2\theta)$. This is a lemniscate (figure-eight curve) centered at the pole with loops oriented along the polar axis and maximum reach $r = 2$ at.

13 [Intermediate] • Target: Permutations & Combinations Practice Exercises

A block of mass $4\ \text{kg}$ is pulled along a rough horizontal plane ($\mu = 0.25$) by a pulling force of magnitude $25\ \text{N}$ acting at 30° above the horizontal. Taking $g = 9.8\ \text{m/s}^2$: (a) show that the normal reaction $R = 26.7\ \text{N}$, (b) find the acceleration of the block.

(a) Resolve vertically (equilibrium in vertical direction): $R + 25\sin 30^\circ = mg \Rightarrow R + 25(0.5) = 4(9.8) \Rightarrow R + 12.5 = 39.2 \Rightarrow R = 26.7\ \text{N}$ (b) Limiting friction force opposes motion: $F = \mu R = 0.25 \times 26.7 = 6.675\ \text{N}$ Resolve horizontally using $F_{\text{net}} = ma$: $25\cos 30^\circ - F = ma \Rightarrow 25\left(\frac{\sqrt{3}}{2}\right) - 6.675 = 4a$.

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14 [Intermediate] • Target: Permutations & Combinations Practice Exercises

Two particles P of mass 3 kg and Q of mass 5 kg are connected by a light inextensible string passing over a smooth fixed pulley. The system is held at rest and released. Taking $g = 9.8$ m/s^2: (a) find the acceleration a of the system, (b) calculate the tension T in the string, (c) find the force exerted by the string on the pulley.

(a) Write $F = ma$ for each particle in its direction of motion: For Q (moving downwards): $5g - T = 5a$ For P (moving upwards): $T - 3g = 3a$ Add the equations: $2g = 8a \Rightarrow a = \frac{2(9.8)}{8} = 2.45$ m/s^2 (b) Substitute a into the equation for P: $T = 3g + 3a = 3(9.8 + 2.45) = 3(12.25) = 36.75$ N ≈ 36.8 N (c) The pulley supports two vertical segments of string with tension T, so the total downward force on the pulley is.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Permutations & Combinations Practice Exercises • Cross-Lesson: 7.01 Differentiation and Rates of Change

A continuous random variable X has PDF given by $f(x) = \frac{k}{x^2}$ for $1 \le x \le 4$, and 0 elsewhere. (a) Show that $k = \frac{4}{3}$. (b) Find the median of X.

(a) Integrate over the support and equate to 1: $\int_1^4 \frac{k}{x^2}\,dx = k\left[-\frac{1}{x}\right]_1^4 = k\left(-\frac{1}{4} - (-1)\right) = \frac{3}{4}k = 1 \Rightarrow k = \frac{4}{3}$ (b) Find the CDF for $1 \le x \le 4$: $F(x) = \int_1^x \frac{4}{3t^2}\,dt = \frac{4}{3}\left(1 - \frac{1}{x}\right)$ Set $F(m) = 0.5$ to find the median m.

16 [Challenging] • Target: Permutations & Combinations Practice Exercises • Cross-Lesson: 8.01 Integration and Kinematics

Decompose the rational expression $R(x) = \frac{7x - 1}{x^2 - x - 6}$ into partial fractions using the Heaviside cover-up method.

Step 1: Factor denominator: $x^2 - x - 6 = (x - 3)(x + 2)$. Set up the decomposition: $\frac{7x - 1}{(x - 3)(x + 2)} = \frac{A}{x - 3} + \frac{B}{x + 2}$. Step 2: Apply Heaviside cover-up method:For A (evaluate at $x = 3$): $A = \frac{7x - 1}{x + 2}\Big|_{x = 3} = \frac{7(3) - 1}{3 + 2} = \frac{20}{5} = 4$. For B (evaluate at $x = -2$): $B = \frac{7x - 1}{x - 3}\Big|_{x = -2} = \frac{7(-2) - 1}{-2 - 3} = \frac{-15}{-5} = 3$. The partial fraction decomposition is $\frac{4}{x - 3} + \frac{3}{x + 2}$.

17 [Challenging] • Target: Permutations & Combinations Practice Exercises • Cross-Lesson: 1.01 Functions, Modulus and Quadratics

Write the standard equation of the ellipse having foci at $(\pm 3, 0)$ and vertices at $(\pm 5, 0)$. State the lengths of the major and minor axes, and compute the eccentricity.

Step 1: Identify ellipse parameters: The vertices and foci lie on the x-axis, so the major axis is horizontal and centered at $(0, 0)$. Here semi-major axis $a = 5$ and focal distance $c = 3$. Step 2: Compute semi-minor axis b: For an ellipse: $b^2 = a^2 - c^2 = 5^2 - 3^2 = 25 - 9 = 16 \Rightarrow b = 4$. Step 3: Formulate equation and invariants: $\frac{x^2}{25} + \frac{y^2}{16} = 1$. Major axis length $2a = 10$, minor axis length $2b = 8$, and eccentricity $e = \frac{c}{a} = \frac{3}{5} = 0.6$.

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18 [Challenging] • Target: Permutations & Combinations Practice Exercises • Cross-Lesson: 2.01 Simultaneous Equations and Inequalities

Find the 6th term in the expansion of $(3x - 2y^2)^8$ in fully simplified form.

In the binomial theorem expansion $\sum_{k=0}^{n} \binom{n}{k} A^{n-k} B^k$, the 1st term has $k = 0$, so the 6th term has $k = 5$. Identify terms: $n = 8$, $A = 3x$, $B = -2y^2$: $T_6 = \binom{8}{5}(3x)^{8-5}(-2y^2)^5 = \binom{8}{3}(3x)^3(-2y^2)^5$. Compute each factor: $\binom{8}{3} = \frac{8 \times 7 \times 6}{3 \times 2 \times 1} = 56 (3x)^3 = 27x^3 (-2y^2)^5 = -32y^{10}$ Multiply the coefficients:

$T_6 = 56 \times 27 \times (-32) x^3 y^{10} = 56 \times (-864) x^3 y^{10} = -48384 x^3 y^{10}$.

19 [Challenging] • Target: Permutations & Combinations Practice Exercises • Cross-Lesson: 3.01 Indices, Surds and Polynomials

The number of accidents on a highway follows a Poisson distribution with mean 1.6 per week. (a) Find the probability of at least 3 accidents in a 2-week period. (b) Given that at least one accident occurs in a week, find the probability that there are exactly 2 accidents in that week.

(a) In a 2-week period, the parameter scales to $\lambda = 1.6 \times 2 = 3.2$:

$P(X \geq 3) = 1 - P(X \leq 2) = 1 - e^{-3.2}\left(1 + 3.2 + \frac{3.2^2}{2}\right) = 1 - 9.32e^{-3.2} \approx 1 - 0.3799 = 0.620$ (3 s. f.) (b) For 1 week, $\lambda = 1.6$. Use conditional probability:

$P(X = 2 \mid X \geq 1) = \frac{P(X=2)}{P(X \geq 1)} = \frac{\frac{e^{-1.6}(1.6)^2}{2}}{1 - e^{-1.6}} = \frac{1.28e^{-1.6}}{1 - e^{-1.6}} \approx \frac{0.2584}{0.7981} \approx 0.324$ (3 s. f.)

20 [Challenging] • Target: Permutations & Combinations Practice Exercises • Cross-Lesson: 4.01 Logarithmic and Exponential Functions

Find the equation of the perpendicular bisector of the line segment joining $P(2, -1)$ and $Q(8, 7)$. Give your answer in the form $y = mx + c$.

Step 1: Find the midpoint M of PQ: $M = \left(\frac{2+8}{2}, \frac{-1+7}{2}\right) = (5, 3)$ Step 2: Find the gradient of line PQ: $m_{PQ} = \frac{7-(-1)}{8-2} = \frac{8}{6} = \frac{4}{3}$ Step 3: Determine the perpendicular gradient: $m_{\perp} = -\frac{3}{4}$ Step 4: Form the equation passing through $M(5, 3)$: $y - 3 = -\frac{3}{4}(x - 5) \Rightarrow y - 3 = -\frac{3}{4}x + \frac{15}{4}$
$y = -\frac{3}{4}x + \frac{27}{4}$ or $y = -0.75x + 6.75$

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Permutations & Combinations Practice Exercises

Evaluate the number of permutations \({}^{5}P_{2} \).

a. 25

b. 21

c. 10

d. **20** ✓

\({}^{n}P_{r} = \frac{n!}{(n-r)!} \implies {}^{5}P_{2} = 5 \times 4 = 20 \).

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22 [Foundational] • Target: Permutations & Combinations Practice Exercises

A committee of 3 people is to be selected from a group of 5 people. The number of ways to form this committee is given by the combination formula ${}^{n}C_{r} = \frac{n!}{r!\,(n-r)!}$. How many different committees can be formed?

a. 10 ✓

b. 15

c. 20

d. 60

The problem requires us to find the number of combinations of selecting 3 people from a group of 5. This is a direct application of the combination formula ${}^{n}C_{r}$, where n is the total number of items to choose from, and r is the number of items to choose. In this case, $n = 5$ and $r = 3$. Using the formula: ${}^{5}C_{3} = \frac{5!}{3!\,(5-3)!} = \frac{5!}{3!\,2!} = \frac{5 \times 4 \times 3!}{3! \times (2 \times 1)} = \frac{5 \times 4}{2} = \frac{20}{2} = 10$. Therefore, there are 10 different committees that can be forme.

23 [Foundational] • Target: Permutations & Combinations Practice Exercises

In how many distinct ways can 6 different people be seated around a circular table (rotations considered identical)?

a. 120 ✓

b. 103

c. 122

d. 100

Applying Permutations & Combinations principles, $(n-1)! = (6-1)! = 120$ Therefore, the verified correct option is 120.

24 [Foundational] • Target: Permutations & Combinations Practice Exercises

Find the number of distinct arrangements of the letters of the word PUPPY.

a. 103

b. 27

c. 64

d. 60 ✓

Applying Permutations & Combinations principles, $\frac{5!}{3!} = 60$ Therefore, the verified correct option is 60.

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25 [Foundational] • Target: Permutations & Combinations Practice Exercises

In how many ways can 8 different books be arranged in a row on a shelf?

a. 45808

b. 35108

c. 40625

d. **40320** ✓

Applying Permutations & Combinations principles, $8! = 40320$ Therefore, the verified correct option is 40320.

26 [Foundational] • Target: Permutations & Combinations Practice Exercises

In how many ways can 6 different people be seated in a row of 6 chairs?

a. **720** ✓

b. 817

c. 800

d. 725

Applying Permutations & Combinations principles, $6! = 720$ Therefore, the verified correct option is 720.

27 [Foundational] • Target: Permutations & Combinations Practice Exercises

In how many ways can 11 different people be arranged in a row if two particular people must NOT sit next to each other?

a. 32806546

b. 33364823

c. 34615086

d. **32659200** ✓

Total − together $= 11! - (11-1)! \times 2! = 39916800 - 7257600 = 32659200$

28 [Foundational] • Target: Permutations & Combinations Practice Exercises

In how many ways can 8 different people be arranged in a row if two particular people must sit together?

a. **10080** ✓

b. 8546

c. 9551

d. 9288

Treat the pair as one unit: $(8-1)! \times 2! = 5040 \times 2 = 10080$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Permutations & Combinations Practice Exercises

In how many ways can 6 different runners be lined up at the start of a race if two particular runners must stand next to each other?

a. 273

b. **240** ✓

c. 220

d. 241

Treat the pair as one unit: $(6-1)! \times 2! = 120 \times 2 = 240$

30 [Intermediate] • Target: Permutations & Combinations Practice Exercises

In how many ways can 5 of 7 different people be selected and arranged in a row?

a. 2891

b. 2282

c. **2520** ✓

d. 2348

${}^{7}P_{5} = \frac{7!}{(7-5)!} = 2520$

31 [Intermediate] • Target: Permutations & Combinations Practice Exercises

The histogram below shows the distribution of heights of a group of students. The total number of students in the group is 50. If the frequency density for the class interval $60 \leq h < 70$ is 1.2, find the value of k.

a. 10

b. **12** ✓

c. 15

d. 20

The area of each bar in a histogram represents the frequency of the corresponding class interval. The frequency density is given by $\frac{\text{frequency}}{\text{class width}}$. The class interval $60 \leq h < 70$ has a class width of $70 - 60 = 10$. The frequency density for this interval is given as 1.2. Therefore, the frequency for this interval is frequency density $\times$ class width $= 1.2 \times 10 = 12$. The histogram shows that the frequency for the interval $60 \leq h < 70$ is represented by k..

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32 [Intermediate] • Target: Permutations & Combinations Practice Exercises

In triangle ABC, angle ABC = θ degrees, and angle ACB = 2θ degrees. The lengths of sides AB and AC are 10 cm and 15 cm respectively. The length of BC is denoted by x cm. Given that the area of triangle ABC is 60 cm^2, find the value of x, to 3 significant figures.

a. 13.7

b. 11.4

c. **12.8** ✓

d. 14.9

Let the angles of triangle ABC be denoted as follows: $\angle ABC = \theta$, $\angle ACB = 2\theta$. By the sine rule, $\frac{AC}{\sin(\theta)} = \frac{AB}{\sin(2\theta)}$. Substituting the given values, $\frac{15}{\sin(\theta)} = \frac{10}{\sin(2\theta)}$. Using the double angle identity, $\sin(2\theta) = 2\sin(\theta)\cos(\theta)$, we get $\frac{15}{\sin(\theta)} = \frac{10}{2\sin(\theta)\cos(\theta)}$. Since $\sin(\theta) eq 0$ for a non-degenerate triangle, we can cancel $\sin(\theta)$ from both sides.

33 [Intermediate] • Target: Permutations & Combinations Practice Exercises

The diagram shows a curve with equation $y = ax^2 + bx + c$ and a straight line with equation $y = mx + d$. The curve and the line intersect at points A and B. The vertex of the parabola is at point V. The x-coordinates of the points A, B, and V are given as $x_A = -1$, $x_B = 3$, and $x_V = 1$ respectively. The y-coordinate of the vertex V is $y_V = -4$. Determine the value of a.

a. **1** ✓

b. 2

c. -1

d. -2

The vertex of a parabola $y = ax^2 + bx + c$ is located at $x = -\frac{b}{2a}$. We are given that the x-coordinate of the vertex V is $x_V = 1$. Therefore, $1 = -\frac{b}{2a}$, which implies $b = -2a$. The parabola passes through its vertex V at $(1, -4)$. Substituting these coordinates into the parabola equation: $-4 = a(1)^2 + b(1) + c$ $-4 = a + b + c$. Substituting $b = -2a$: $-4 = a + (-2a) + c$ $-4 = -a + c$ (Equation 1) The parabola and the line intersect at points A and B, with x-coordinates $x_A = -1$ and.

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34 [Intermediate] • Target: Permutations & Combinations Practice Exercises

In triangle ABC, angle ABC = 90°, AB = 10 cm, and angle BAC = θ. Given that $\sin\theta = \frac{3}{5}$, find the length of AC.

a. 16.7 cm

b. 8.0 cm

c. **12.5 cm ✓**

d. 6.0 cm

We are given a right-angled triangle ABC, where angle ABC = 90°. We know that AB = 10 cm and angle BAC = θ. We are also given that $\sin\theta = \frac{3}{5}$. In a right-angled triangle, the sine of an angle is defined as the ratio of the length of the side opposite the angle to the length of the hypotenuse. Therefore, $\sin\theta = \frac{\text{Opposite}}{\text{Hypotenuse}}$. In triangle ABC, the side opposite to angle θ (angle BAC) is BC, and the hypotenuse is AC. So.

35 [Intermediate] • Target: Permutations & Combinations Practice Exercises

A circle is centered at the origin and passes through the point $P(a, 4)$ in the first quadrant. The equation of a tangent line to the circle at point P is given by $3x + 4y = k$. Find the value of k.

a. **25 ✓**

b. 30

c. 35

d. 40

The equation of a circle centered at the origin is $x^2 + y^2 = r^2$. Since point $P(a, 4)$ lies on the circle, we have $a^2 + 4^2 = r^2$, so $a^2 + 16 = r^2$. The radius from the origin to $P(a, 4)$ is perpendicular to the tangent line at P. The slope of the radius is $m_{radius} = \frac{4-0}{a-0} = \frac{4}{a}$. The given tangent line is $3x + 4y = k$, which can be rewritten in slope-intercept form as $y = -\frac{3}{4}x + \frac{k}{4}$. The slope of the tangent line is $m_{tangent} = -\frac{3}{4}$. Since the rad.

36 [Intermediate] • Target: Permutations & Combinations Practice Exercises

The graph of the quadratic function $y = x^2 - 4x + 3$ is shown. The graph of the linear function $y = x - 1$ is also shown. The graphs intersect at points A and B. Find the coordinates of point A.

a. **(1, 0) ✓**

b. (3, 2)

c. (0, 3)

d. (2, 1)

To find the intersection points, we set the two equations equal to each other: $x^2 - 4x + 3 = x - 1$. Rearranging the terms to form a quadratic equation: $x^2 - 5x + 4 = 0$. Factoring the quadratic equation: $(x - 1)(x - 4) = 0$. This gives us two possible x-values for the intersection points: $x = 1$ or $x = 4$. For $x = 1$, we can find the corresponding y-coordinate using the linear equation $y = x - 1$: $y = 1 - 1 = 0$. So, one intersection point is $(1, 0)$. For $x = 4$, we can find the corresponding y-coordinate.

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37 [Intermediate] • Target: Permutations & Combinations Practice Exercises

The diagram shows part of the curve with equation $y = x^2 - 4x + 5$ and the point $P(3, 2)$ lying on the curve. Find the gradient of the tangent to the curve at the point P.

a. 2 ✓

b. 6

c. -4

d. $\frac{2}{3}$

To find the gradient of the tangent to the curve at $P(3, 2)$, we determine the derivative $\frac{dy}{dx}$ and evaluate it at $x = 3$. 1. Differentiate $y = x^2 - 4x + 5$ with respect to x using the power rule: $\frac{dy}{dx} = 2x - 4$ 2. Substitute $x = 3$ into the derivative: $\frac{dy}{dx}\Big|_{x=3} = 2(3) - 4 = 6 - 4 = 2$ Common errors: - 6 occurs if only the first term is differentiated (giving $2x$) and evaluated at $x = 3$, omitting the deriv.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Permutations & Combinations Practice Exercises • Cross-Lesson: 2.01 Simultaneous Equations and Inequalities

In triangle ABC, angle A is 52 degrees. The side opposite angle B (AC) has a length of 10 units, and the side opposite angle C (AB) has a length of 14 units. What is the length of the side opposite angle A (BC)?

a. 10.3

b. **12.1** ✓

c. 14.8

d. 8.5

We can use the Law of Cosines to solve for the unknown side BC. The Law of Cosines states that for any triangle with sides a, b, and c, and opposite angles A, B, and C respectively: $a^2 = b^2 + c^2 - 2bc\cos(A)$. In this problem, we have: side b = AC = 10, side c = AB = 14, and angle A = 52 degrees. We want to find side a = BC. Substituting the values into the formula: $a^2 = 10^2 + 14^2 - 2(10)(14)\cos(52^\circ)$ $a^2 = 100 + 196 - 280\cos(52^\circ)$ $a^2 = 296 - 280(0.61566)$ $a^2 = 296 - 172.3848$.

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39 [Challenging] • Target: Permutations & Combinations Practice Exercises • Cross-Lesson: 3.01 Indices, Surds and Polynomials

In the right-angled triangle ABC, $\angle ABC = 90^\circ$. If $AB = 5$ cm and $BC = 12$ cm, find the length of the hypotenuse AC.

a. 10 cm

b. **13 cm ✓**

c. 17 cm

d. 144 cm

The triangle ABC is a right-angled triangle with $\angle ABC = 90^\circ$. We are given the lengths of the two shorter sides, AB = 5 cm and BC = 12 cm. We can use the Pythagorean theorem to find the length of the hypotenuse AC. The Pythagorean theorem states that in a right-angled triangle, the square of the hypotenuse (the side opposite the right angle) is equal to the sum of the squares of the other two sides. $AC^2 = AB^2 + BC^2$. Substituting the given values: $AC^2 = 5^2 + 12^2$. $AC^2 = 25 + 144$. $AC^2 = 169$.

40 [Challenging] • Target: Permutations & Combinations Practice Exercises • Cross-Lesson: 4.01 Logarithmic and Exponential Functions

A circle has a radius of 6 units. A chord within this circle subtends a sector with a central angle of 60 degrees. What is the length of the chord?

a. **6 units ✓**

b. $6\sqrt{3}$ units

c. 12 units

d. $12\sqrt{3}$ units

We have a circle with radius $r = 6$ and a sector with a central angle of 60 degrees. The chord connecting the endpoints of the arc of this sector forms an isosceles triangle with the two radii. Since the central angle is 60 degrees and the other two angles in the triangle must be equal (because the sides opposite them are radii), each of the other angles is $(180^\circ - 60^\circ)/2 = 60^\circ$. Therefore, the triangle is equilateral. In an equilateral triangle, all sides are equal. Since two sides are radii of.

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41 [Challenging] • Target: Permutations & Combinations Practice Exercises • Cross-Lesson: 5.01 Trigonometric Functions and Circular Measure

The diagram shows a right-angled triangle ABC, with the right angle at B. The length of side AB is 10 cm and the length of side BC is 24 cm. A semicircle is drawn with AC as its diameter. Calculate the area of the semicircle in terms of π.

a. 169π cm^2

b. 84.5π cm^2 ✓

c. 13π cm^2

d. 169 cm^2

First, we need to find the length of the hypotenuse AC using the Pythagorean theorem: $AC^2 = AB^2 + BC^2$. $AC^2 = 10^2 + 24^2 = 100 + 576 = 676$. So, $AC = \sqrt{676} = 26$ cm. The radius of the semicircle is half the diameter AC, so the radius $r = \frac{26}{2} = 13$ cm. The area of a semicircle is given by $\frac{1}{2}\pi r^2$. Substituting the radius, we get $\frac{1}{2}\pi(13)^2 = \frac{1}{2}\pi(169) = 84.5\pi$ cm^2. Common error: Using the sides AB or BC as diameter, or for.

42 [Challenging] • Target: Permutations & Combinations Practice Exercises • Cross-Lesson: 7.01 Differentiation and Rates of Change

The diagram shows a number line. The distance between the points marked A and B is 15 units. The coordinate of A is -7. Find the possible coordinates of B.

a. **-22 and 8** ✓

b. 8 and 22

c. -22 and -8

d. 7 and 22

Let the coordinate of point B be x. The distance between two points on a number line is the absolute difference of their coordinates. Therefore, the distance between A and B is $|x - (-7)| = |x + 7|$. We are given that this distance is 15 units. So, we have the equation $|x + 7| = 15$. This gives us two possibilities: $x + 7 = 15$ or $x + 7 = -15$. Solving the first equation, $x = 15 - 7 = 8$. Solving the second equation, $x = -15 - 7 = -22$. Therefore, the possible coordinates of B are -22 and 8.

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43 [Challenging] • Target: Permutations & Combinations Practice Exercises • Cross-Lesson: 8.01 Integration and Kinematics

The diagram shows the graph of the quadratic function $y = x^2 + bx + c$. The graph intersects the x-axis at $(1, 0)$ and $(5, 0)$. Find the values of b and c.

a. $b = -6, c = 5$ ✓

b. $b = 6, c = 5$

c. $b = -6, c = -5$

d. $b = 6, c = -5$

Since the graph intersects the x-axis at $(1, 0)$ and $(5, 0)$, these are the roots of the quadratic equation $x^2 + bx + c = 0$. We can use Vieta's formulas, which state that for a quadratic equation $ax^2 + bx + c = 0$, the sum of the roots is $-b/a$ and the product of the roots is c/a. In this case, $a = 1$. Sum of roots: $1 + 5 = 6$. Therefore, $-b/1 = 6$, which means $b = -6$. Product of roots: $1 \times 5 = 5$. Therefore, $c/1 = 5$, which means $c = 5$. Thus, the values are $b = -6$ and $c = 5$.

44 [Challenging] • Target: Permutations & Combinations Practice Exercises • Cross-Lesson: 1.01 Functions, Modulus and Quadratics

In a right-angled triangle, one of the acute angles is 30° and the side opposite to this angle is 6 cm. Calculate the length of the hypotenuse.

a. 10 cm

b. **12 cm** ✓

c. 8 cm

d. 6 cm

Let the right-angled triangle be ABC, with the right angle at C. Let angle A be 30°. The side opposite to angle A is BC, which is given as 6 cm. The hypotenuse is AB. We use the sine function: $\sin(A) = \frac{\text{opposite}}{\text{hypotenuse}}$. So, $\sin(30^\circ) = \frac{BC}{AB}$. We know that $\sin(30^\circ) = \frac{1}{2}$. Therefore, $\frac{1}{2} = \frac{6}{AB}$. Solving for AB, we get $AB = 2 \times 6 = 12$ cm.

45 [Challenging] • Target: Permutations & Combinations Practice Exercises • Cross-Lesson: 2.01 Simultaneous Equations and Inequalities

In a right-angled triangle ABC, where the right angle is at C, if $AC = 10$ and $BC = 24$, find the length of the hypotenuse AB.

a. 25

b. **26** ✓

c. 27

d. 28

We can use the Pythagorean theorem to find the length of the hypotenuse in a right-angled triangle. The theorem states that $a^2 + b^2 = c^2$, where a and b are the lengths of the two shorter sides (legs) and c is the length of the hypotenuse. In this triangle, $AC = 10$ and $BC = 24$. Therefore, we have: $AB^2 = AC^2 + BC^2$ $AB^2 = 10^2 + 24^2$ $AB^2 = 100 + 576$ $AB^2 = 676$ $AB = \sqrt{676}$ $AB = 26$ The length of the hypotenuse AB is 26.

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